

Quantitative Spatial Economics for Policy Evaluation

Topic 0: Course context and roadmap

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CEMFI Summer School 2026

31 August–4 September 2026

Why quantitative spatial economics?

- ▶ Economic activity, opportunity, and exposure to policy are unevenly distributed across space.
- ▶ Transport, housing, land-use, environmental, and place-based policies change several markets simultaneously.
- ▶ Quantitative spatial models connect microeconomic behaviour, geography, data, and equilibrium adjustment.
- ▶ They are increasingly used as workhorse tools for policy design and appraisal.

Central question: How does a geographically targeted intervention propagate through commuting, wages, employment, rents, land use, and welfare?

From measurement to policy appraisal

1. Specify workers, firms, locations, mobility, and spatial frictions.
2. Discipline preferences, technologies, and frictions using observed data.
3. Recover unobserved local fundamentals consistent with the observed equilibrium.
4. Change a policy-relevant primitive.
5. Solve the new spatial equilibrium and compare outcomes and welfare.

This sequence makes assumptions transparent and forces internal consistency, but does not eliminate the need for credible empirical evidence and careful validation.

Policy evaluation in general equilibrium

1. **Map the policy into a primitive.** Identify the exogenous model object that the intervention directly changes—for example a bilateral travel cost, tax, regulation, amenity, productivity shifter, or land-supply constraint.
2. **Re-solve the general equilibrium.** Let that primitive change propagate jointly through all endogenous outcomes in interacting goods and factor markets.

The discipline of a quantitative counterfactual

The experiment changes a primitive that corresponds to a defensible exogenous policy intervention. It does not impose a change in an endogenous outcome such as employment, wages, or rents.

The model then determines the effects on prices, quantities, location choices, commuting, land use, income, and welfare while respecting market clearing and equilibrium feedbacks.

Our wider research context

- ▶ The DFG Research Unit on *Growth and Spatial Inequalities* studies how economic growth and structural change interact with disparities across places.
- ▶ Our work combines reduced-form evidence with quantitative models that make spatial mechanisms and policy counterfactuals explicit.
- ▶ A related Department for Transport project develops quantitative approaches to transport appraisal.
- ▶ Transport is a natural application: a network intervention changes bilateral access, location choices, and equilibrium outcomes.

The full QSE curriculum

The material originates in a semester-long PhD course taught at the Berlin School of Economics and Humboldt University.

Neoclassical spatial equilibrium $\longrightarrow$ quantitative urban models $\longrightarrow$ urban models with commuting $\longrightarrow$ quantitative spatial models

The twelve-topic sequence runs from Rosen–Roback and Glaeser–Gottlieb, through Ahlfeldt–Barr, to Ahlfeldt–Redding–Sturm–Wolf, Redding–Rossi–Hansberg, and Monte–Redding–Rossi–Hansberg.

What the full-semester course involves

- ▶ Twelve linked lecture topics, from analytical foundations to large quantitative models.
- ▶ Extensive weekly problem sets.
- ▶ Regular programming and construction of numerical solution algorithms.
- ▶ Work with replication directories, data, and computational toolkits.
- ▶ A cumulative learning process in which later exercises build on earlier code.

The full format develops computational independence; it cannot be compressed into one summer-school week without changing the learning model.

The focused CEMFI route

We select five topics and follow one coherent line:

1. Rosen–Roback and the quantitative Glaeser–Gottlieb framework.
2. Discrete choice and preference heterogeneity.
3. The Ahlfeldt–Redding–Sturm–Wolf model.
4. Quantification of its primitives and fundamentals.
5. Counterfactual policy evaluation.

Lectures go deeper into the theory behind this sequence. Tutorials are necessarily more guided and use an existing computational infrastructure.

Lectures and practical sessions

Day	Lecture	Practical session (15:00–17:00)
Monday	Topic 1	—
Tuesday	Topic 5	—
Wednesday	Topic 6: model	Toolkit setup
Thursday	Topic 7: quantification	Reproduce quantification
Friday	Topic 8: counterfactuals	Reproduce counterfactuals

The teaching-assistant sessions guide you through the code step by step. The objective is to understand how theoretical objects map into computational objects, not merely to press *Run*.

Optional capstone: transport appraisal

- ▶ Construct travel-time matrices for two Berlin transport networks.
- ▶ Treat the complete network as the with-U5 scenario.
- ▶ Remove the full Alexanderplatz–Hauptbahnhof U-Bahn connection for the comparison network.
- ▶ Translate the accessibility change into the ARSW model and solve the equilibrium response.
- ▶ Separate direct travel-time effects from endogenous changes in employment, wages, rents, and welfare.

The capstone is independent and optional. It connects GIS/network construction to quantitative spatial policy evaluation.

What this week can—and cannot—deliver

By Friday, you should be able to

- ▶ read the equilibrium structure;
- ▶ understand the quantification logic;
- ▶ navigate the toolkit;
- ▶ interpret counterfactual output critically.

Beyond this week

- ▶ write solvers from scratch;
- ▶ develop new identification strategies;
- ▶ build production-ready policy models;
- ▶ undertake extensive robustness analysis.

Continue with the full experience

Students interested in the complete semester-long experience are welcome to visit the Berlin School of Economics and Humboldt University.

Visiting students have regularly joined the course, worked through the full problem-set sequence, and participated in the wider quantitative spatial economics community in Berlin.

www.bqse.de

Participation is generally possible without a course fee; practical arrangements and current availability should be confirmed in advance.

Roadmap for the week

Spatial equilibrium $\rightarrow$ heterogeneous preferences $\rightarrow$ urban model
with commuting $\rightarrow$ quantification $\rightarrow$ counterfactuals

Recurring discipline

At every stage, distinguish assumptions and primitives from equilibrium outcomes, and distinguish model-based counterfactuals from causal estimates.